Master of Engineering Information Technology · Individual Project

Intelligent Ultrasonic Sensing for Movement Intensity Classification using Hybrid Machine Learning

Mandar Gokul Kale

Matriculation No. 1501517

Guided by Dr. Andreas Pech

sensing question

Ultrasound detects people without a camera, but accuracy can mislead

1. **Camera free sensing:** Ultrasonic sensing detects presence and activity from reflected acoustic signals, without a camera.
2. **Two tasks:** human presence detection (human vs EmptyRoom/ChairTable) and Standing vs Reading.
3. **Positional cues:** Distance and amplitude can act as unwanted positional cues, so position aware and position normalized representations are compared.
4. **Goal:** understand how signal representation, model, environment and subject affect performance, including cross environment generalization.

prior work · research gap

Prior work leaves a gap in representation, normalization and environment



Cameras raise privacy issues; PIR, radar and ultrasonic trade off accuracy, cost and privacy.



Aggregate accuracy hides condition specific failures.

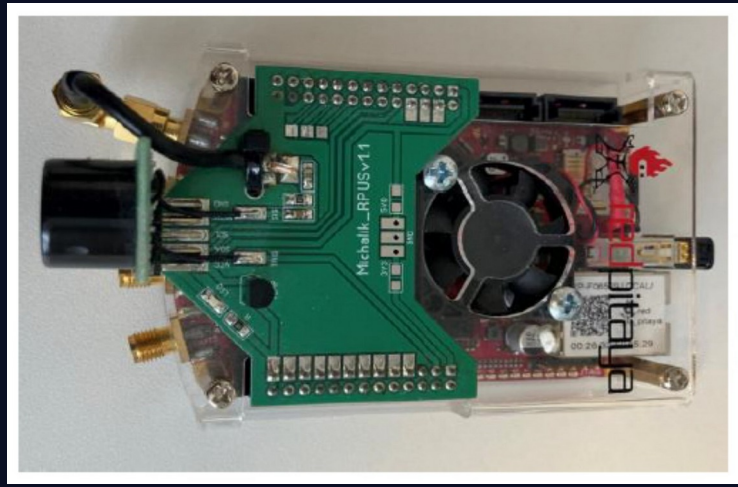


Representation effect is rarely separated from model family.



Non-human conditions are treated as one class, though they differ in difficulty.

Red Pitaya STEMlab platform used for ultrasonic signal acquisition



acquisition · fixed geometry

Red Pitaya digitizes SRF02 echoes at about 1.95 MS/s

01

SRF02 transducer

About 40 kHz pings, triggered over I2C.

02

Red Pitaya STEMlab

14-bit DMA acquisition; decimation 64 from 125 MS/s.

03

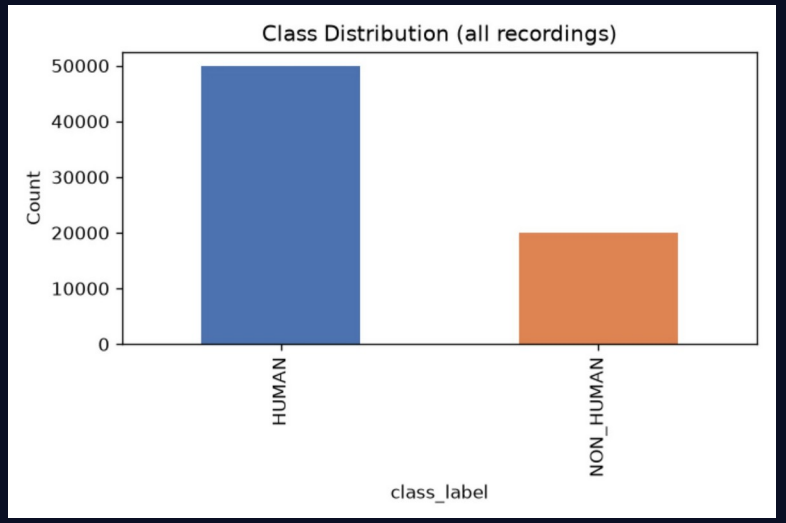
Laptop

25,000 samples per frame, sent by UDP.

sensor on tripod · about 2.1 m · fixed geometry

Four conditions, two environments, 70,000 recordings

- **EmptyRoom:** background baseline, no person
- **ChairTable:** static furniture; tests separating objects from humans
- **HumanReading:** seated subject reading
- **HumanStanding:** standing subject
- Conditions analyzed separately, not as one non human class
- Subject level analysis: Person 3 and Person 4 in Env2 (anonymized)



Class distribution: 50,000 HUMAN vs 20,000 NON_HUMAN

Env1: Laboratory · 30,000 recordings · Train and tune only

Env2: Corridor · 40,000 recordings · Test once

Main binary pipeline · 50,000 HUMAN vs 20,000 NON_HUMAN

experimental design

Representation–model combinations were systematically compared

Position treatment

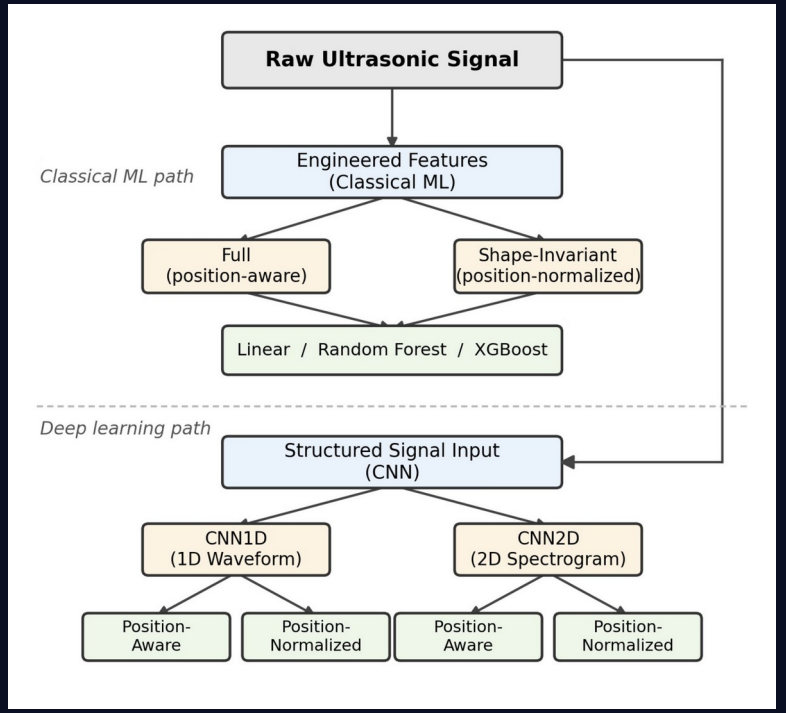
Position aware vs position normalized via Hilbert envelope onset alignment.

Representation → model family

Full or shape invariant features, raw waveform, band limited spectrogram.

Representation	Model family
Full or shape invariant features	Classical: Linear, Random Forest, XGBoost
Raw waveform	CNN1D
Band limited spectrogram	CNN2D

Adam · binary cross-entropy · early stopping · frozen models



Representation taxonomy: each input representation paired with its model family

echo characteristics

Non-human echoes are stronger than human echoes

Consistent spectral peak near 40 kHz in every condition.

Energy sits in a short echo burst against near silence.

RMS order
EmptyRoom
ChairTable
HumanReading
HumanStanding

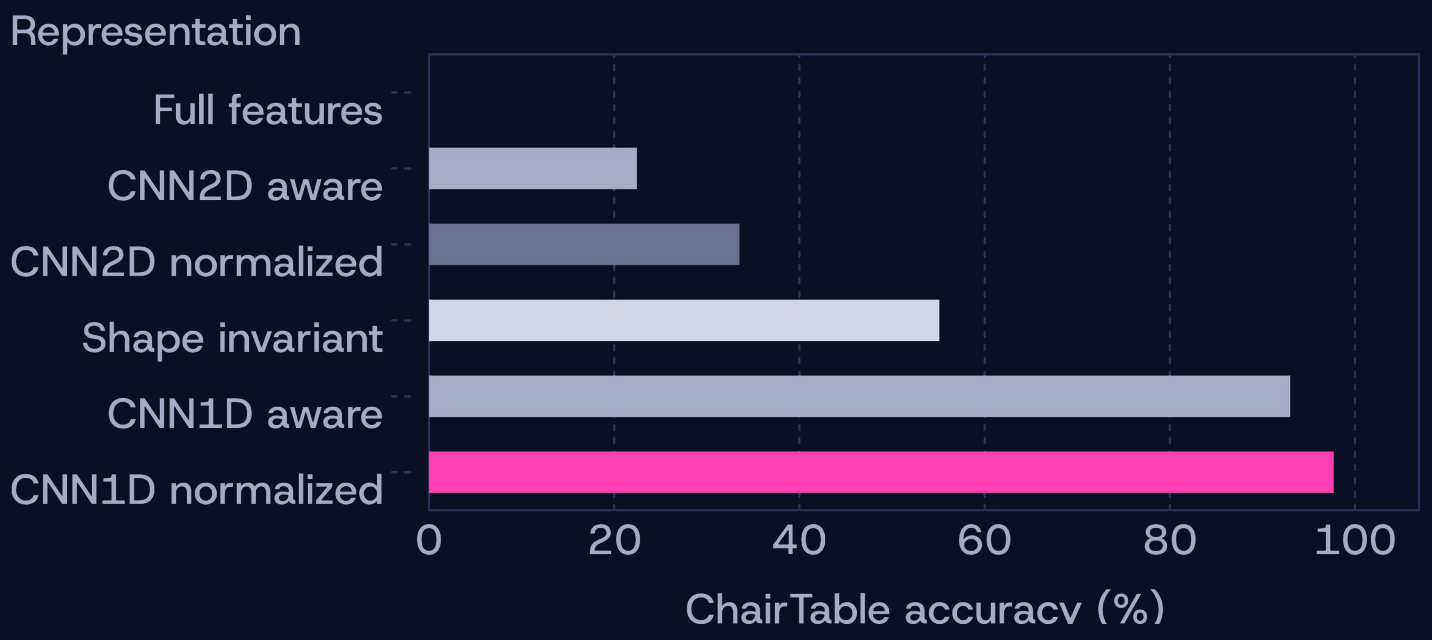
Hard, flat surfaces reflect more coherently than soft human bodies.



Hard, flat reflectors in the echo scene.

Env2 · ChairTable accuracy

ChairTable breaks generalization: 0% to 97.72% by representation



Full-feature models reached 99.93% validation accuracy on Env1, yet labeled every ChairTable recording HUMAN in Env2.

EmptyRoom stayed at 93 to 98%.



ChairTable condition: an empty chair beneath the sensor

normalization · trade-off

Normalization helps ChairTable but costs Reading accuracy

HumanReading falls under every normalized representation

Representation	Position-aware	Position-normalized
Shape	99.51	90.89
CNN1D	98.59	86.06
CNN2D	99.93	96.36

Standing vs Reading

Model	Aware	Normalized
CNN2D	97.13%	96.86%
CNN1D	91.58%	95.30%

No clear benefit from position-aware input: CNN2D 97.13% vs 96.86%; CNN1D 91.58% vs 95.30%. Subject effect: Person 3 outperformed Person 4 across all representations.

HumanReading condition: a seated subject reading beneath the sensor



Conclusion

1. **Aggregate accuracy overstates reliability:** ChairTable ranges from 0.00% (full hand crafted features) to 97.72% (position normalized CNN1D).
2. **CNN1D with position-normalized raw waveforms** was the strongest configuration for cross-environment human-presence generalization, ahead of classical models and band limited CNN2D.
3. **Standing vs Reading:** no clear benefit from position-aware input (CNN2D 97.13% aware, 96.86% normalized) normalization cost HumanReading accuracy.
4. **Future work:** CNN1D filter visualization, remove only timing or only amplitude, test Person 5 and new environments.

Thank you

Questions?

Mandar Gokul Kale, mandar.kale@stud.fra-uas.de